

Research Statement

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Overview

My research is in number theory, spanning arithmetic geometry, quadratic forms over number fields and arithmetic statistics, with computation as a unifying thread. A recurring theme throughout my research is the passage from qualitative finiteness to explicit arithmetic. Given a theorem asserting that only finitely many objects of a certain kind exist, can one determine them all, understand their structure, and make the resulting data publicly available for further study? The same question arises for asymptotic predictions: can the constants they involve be computed and tested? My work can be organised into the following three themes:

1. **Effective Shafarevich problems:** classifying abelian varieties, especially abelian surfaces over $\mathbb{Q}$, with good reduction outside a fixed small set of primes. This was the subject of my PhD thesis [45].
2. **Quadratic lattices and units over totally real fields:** universal and n -universal lattices, indecomposable elements, sails and criterion sets, and sums of units. This is the main focus of my postdoctoral work in Prague.
3. **Arithmetic statistics:** the distribution of unit signatures and 2-class groups of biquadratic fields, and computing constants in Malle’s conjecture.

Computations run throughout all three themes. I use it not only to check examples, but to formulate conjectures, find exceptional cases and produce data that others can use. I am an Associate Editor of both the LMFDB [41], and the OEIS [31], and I maintain public databases of genus 2 curves, of indecomposables and sails, and of constants in Malle’s conjecture. I have also recently begun formalising the arithmetic theory of quadratic lattices over number fields in Lean 4 [47], with the aim of verifying the results described in Section 2.

1 Effective Shafarevich problems

Let K be a number field, d a positive integer, and S a finite set of places of K . Shafarevich conjectured, and Faltings proved [10], that there are only finitely many K -isomorphism classes of d -dimensional abelian varieties A/K with good reduction outside S . However, Faltings’s proof does not give an effective algorithm for finding all such varieties. This motivates the following problem:

Problem 1 (Effective Shafarevich). *Given a number field K , a positive integer d and a finite set of places S , compute all d -dimensional abelian varieties A/K with good reduction outside S ; equivalently, give an effectively computable bound on their Faltings heights.*

Effective algorithms are known for elliptic curves [8], for hyperelliptic curves [49] and for abelian varieties of GL_2 -type over $\mathbb{Q}$ [50]. However, Problem 1 is open already in the basic case $K = \mathbb{Q}$, $d = 2$, $S = \{2\}$, which in particular contains the following question of Poonen [34, p. 301]:

Problem 2 (Poonen). *Classify all genus 2 curves $C/\mathbb{Q}$ whose Jacobian has good reduction away from 2.*

Merriman and Smart [30, 39] determined the 366 genus 2 curves over $\mathbb{Q}$ that *themselves* have good reduction away from 2. The condition on the Jacobian is weaker, as a curve may have bad reduction at an odd prime p while its Jacobian has good reduction at p . This is what makes Poonen’s problem much harder than the earlier classifications.

1.1 Previous work

A database of genus 2 curves. By combining several search strategies I found many curves beyond the Merriman–Smart list, for example $y^2 = 2x^5 + 956x^3 + 114244x$, which has bad reduction at 13 although its Jacobian is good away from 2. I maintain a database of such genus 2 curves [46], which now contains 512 $\mathbb{Q}$ -isomorphism classes of genus 2 curves with 2-power conductor, as well as 234 isogeny classes of abelian surfaces with good reduction away from 2. For each object, I record models, conductors, Mordell–Weil ranks, torsion, automorphism and Sato–Tate groups, endomorphism algebras and Euler factors.

A complete classification with full 2-torsion. Good reduction away from 2 forces the torsion fields $\mathbb{Q}(A[2^n])$ to be unramified outside 2. When $A[2]$ is rational they are moreover 2-extensions of $\mathbb{Q}$, and this sharply restricts the image of Galois. Combining this with the classification of finite 2-groups, my thesis proves the following theorem:

Theorem 3 ([45, Theorem 4.20]). *There are exactly three isogeny classes of principally polarised abelian surfaces $A/\mathbb{Q}$ with good reduction away from 2 which contain a surface with full rational 2-torsion: those of $E_1 \times E_1$, $E_1 \times E_2$ and $E_2 \times E_2$, where $E_1 : y^2 = x^3 - x$ and $E_2 : y^2 = x^3 - 4x$.*

Curves over cyclotomic $\mathbb{Z}_\ell$ -extensions. With Siksek [38], I studied the analogue of the Shafarevich conjecture over cyclotomic $\mathbb{Z}_\ell$ -extensions $K_{\infty,\ell}/K$, taking S to be the primes above 2 and ℓ . In contrast with the number field case, finiteness fails dramatically. Already the S -unit equation $\varepsilon + \delta = 1$ has infinitely many solutions over $\mathbb{Q}_{\infty,\ell}$ for $\ell \in \{2, 3, 5, 7\}$, even when S is just the prime above ℓ . Moreover, there are infinitely many $K_{\infty,\ell}$ -isomorphism classes of d -dimensional abelian varieties with good reduction outside S for all $\ell \leq 13$, and for all ℓ when d is a multiple of $\lfloor (\ell - 3)/4 \rfloor$.

1.2 Future work

Completing the genus 2 classification: a hybrid method. There are two main approaches to Problem 2. Each has succeeded in special cases, but neither alone resolves it completely.

- *Galois representations.* For an abelian surface $A/\mathbb{Q}$ good away from 2, the images of $\rho_{A,2^n} : G_{\mathbb{Q}} \rightarrow \mathrm{GL}_4(\mathbb{Z}/2^n\mathbb{Z})$ are highly restricted, which constrains the characteristic polynomials of Frobenius modulo 2^n . Via the Faltings–Serre–Livné method [36, 26], this then shows that two such representations agree once their Frobenius data agree at an explicit finite set of primes, so this finite amount of data determines the L-function. This is the method behind Theorem 3.
- *L-functions.* Assuming the expected functional equation, each choice of test function gives, via the approximate functional equation, a linear constraint on the Dirichlet coefficients with rapidly decaying weights. Using interval arithmetic, Farmer, Koutsoliotas and Lemurell [11] used this to find all degree-4 Hasse–Weil L -functions of conductor below 500.

I propose to combine the approaches. First, the Galois-theoretic constraints reduce the Euler factors to a short list of residue classes modulo powers of 2. Next, the analytic constraints are applied to this much smaller search space. Finally, the survivors are matched against explicit Jacobians and products. The new dataset makes this far more practical than when I first began the project, since it provides the known 2-adic images and Euler-factor patterns against which each step can be tested. A first concrete target is to show that the only isogeny class of abelian surfaces over $\mathbb{Q}$ of conductor 1024 is that of $E_1 \times E_1$. This case is beyond either method alone. A working implementation would resolve Problem 2 and serve as a template for other small cases of Problem 1.

Good reduction outside 7 (with P. Goodman). Verhoeck [44] showed, assuming GRH, that an abelian variety over $\mathbb{Q}$ with good reduction outside 7, satisfying a mild ramification condition at 7

(“ramified of level 0 at 7”), is isogenous to a product of copies of $J_0(49)$ and of the abelian surface attached to the newform 49B. GRH enters only through root discriminant bounds, and for abelian surfaces we expect to remove it. More ambitiously, we plan to combine Schoof’s group-scheme method [35] with Faltings–Serre–Livné. The idea is to use restrictions on the 2^n -torsion group schemes to constrain Frobenius polynomials directly, which would give a third ingredient for the hybrid method above. A first step is to classify the possible 2-torsion fields of such surfaces.

Beyond $\mathbb{Q}$ and genus 2. Rather than attacking Problem 1 in general, I aim to find situations where extra structure (prescribed torsion, a constrained endomorphism algebra, few bad primes, low degree base fields) reduces the possible Galois images enough for finite Galois data to separate all isogeny classes. This would give a systematic bridge between explicit class field theory, finite-group computations and Faltings–Serre–Livné arguments. It would also extend the methods above to other sets S , other number fields and hyperelliptic curves of higher genus.

2 Quadratic lattices and units over totally real fields

Let K be a totally real number field with ring of integers $\mathcal{O}_K$ and totally positive elements $\mathcal{O}_K^+$. A totally positive definite $\mathcal{O}_K$ -lattice is *universal* if it represents every element of $\mathcal{O}_K^+$. It is *n -universal* if it represents every totally positive definite $\mathcal{O}_K$ -lattice of rank n . Over $\mathbb{Z}$, universality can be decided by the 15 and 290 theorems [3, 4]: a classical (respectively, integer-valued) positive definite form is universal if and only if it represents a certain explicit finite set of integers, the largest being 15 (respectively 290). Over number fields, the situation is more difficult. Siegel [37] showed that a sum of squares is universal only over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$, and Blomer and Kala [5] showed that the minimal rank of a universal form can be arbitrarily large, even over real quadratic fields. One of the central open problems in this area is *Kitaoka’s conjecture*: only finitely many totally real fields admit a ternary universal form.

The key tool behind recent progress [17] is the set I_K of *indecomposables*, the elements of $\mathcal{O}_K^+$ that cannot be written as the sum of two elements in $\mathcal{O}_K^+$. They are natural obstructions to universal forms of small rank. In real quadratic fields they can be constructed from the continued fraction of $\sqrt{D}$ [9], but in higher degrees there is no simple description of such indecomposables. One promising approach is to study the boundary of the convex hull of $\iota(\mathcal{O}_K^+)$ under the Minkowski embedding ι . This is called the *sail* $\mathcal{S}_K$, which is periodic under the action of units. Kala and Man [19] proved that $\mathcal{S}_K \cap \iota(\mathcal{O}_K) \subseteq \iota(I_K)$, with equality for quadratic fields but not in general.

Another important problem is determining whether a given $\mathcal{O}_K$ -lattice is universal or n -universal. We say a finite set $S \subset \mathcal{O}_K^+$ is a *criterion set* if every form representing all of S is universal. Chan and Oh [7] proved that finite criterion sets always exist. Kala, Krásenský and Romeo [18] characterised the unique minimal one, C_K , and showed that it contains every squarefree indecomposable. However, apart from some small fields, very few cases of a minimal criterion set C_K are known explicitly.

2.1 Previous work

n -universal lattices. Let $U_{\mathcal{O}_K}(n)$ be the minimal rank of an n -universal $\mathcal{O}_K$ -lattice. With Park, Yatsyna and Yoon [33], I proved an explicit asymptotic formula for $\log U_{\mathcal{O}_K}(n)$ as $n \rightarrow \infty$. We also proved effective finiteness results for the fields in which $U_{\mathcal{O}_K}(n)$ is small, bounds on the gaps $U_{\mathcal{O}_K}(n+4) - U_{\mathcal{O}_K}(n)$, and lower bounds on the sizes of minimal n -universal criterion sets. I gave an invited talk on this work at the *Quadratic Lattices in Prague* conference in August 2026.

Sums of two units. Dirichlet’s theorem describes the unit group $\mathcal{O}_K^\times$ completely, yet additive questions about units remain hard. Such problems go back to Nagell and Newman, and several were posed as open problems by Jarden and Narkiewicz [16]. Given a number field K , let N_K be the set of positive integers that are sums of two units of $\mathcal{O}_K$. With Tinková and Yatsyna [42] I showed that N_K is finite if and only if K has no real quadratic subfield, and we solved $\varepsilon + \delta = n$ completely over all cyclic cubic and all complex cubic fields.

Our original finiteness proof bounds the number of elements of N_K but not their size. In a recent paper with Ziegler [48], I proved an effective height bound for elements in N_K by combining the unit matching techniques of Bajpai and Bennett [2] with Matveev’s bounds [29]. Using LLL reduction, we explicitly showed that $N_K = \{1, 2, 3, 4, 5, 6, 7, 10, 11, 15, 251\}$ for the smallest totally real quintic field with Galois group S_5 .

Indecomposables and sails With Kala, Man and Yatsyna I am building a public database [20] of the indecomposables and the fundamental domains of sails of totally real fields. The fields reach well beyond the quadratic and cubic families where indecomposables were previously understood. Since indecomposables have norm bounded by the discriminant [21], this is a finite but very large computation of several CPU-years. It is designed to integrate with the LMFDB, where I have also contributed to the new lattices section [1].

2.2 Future work

Indecomposables and sails in higher degree. Almost everything known about indecomposables and sails concerns mainly quadratic fields and some families of cubic fields [22]. The first results in higher degree [19] already show different behaviour: some biquadratic fields have sails with no face at integer distance greater than 1, contradicting a conjecture of Arnold and Karpenkov. I plan to compute the set of indecomposables I_K and face structure of the sail $\mathcal{S}_K$ systematically for all totally real fields of small degree and bounded discriminant, together with the *deeper sails* $\mathcal{S}_K^{(i)}$ obtained by repeatedly peeling off the outer layer, starting from $\mathcal{S}_K^{(1)} = \mathcal{S}_K$. Since

$$\mathcal{S}_K \cap \iota(\mathcal{O}_K) \subseteq \iota(I_K) \subseteq \mathcal{S}_K^{(1)} \cup \dots \cup \mathcal{S}_K^{(i)}$$

for some depth i , the data will show how large i must be, when $\iota(I_K) = \mathcal{S}_K \cap \iota(\mathcal{O}_K)$ holds, and which face types occur and how often. None of these questions has a concrete known answer beyond degree 2. The same objects exist also in every signature and in non-maximal orders, where even basic questions, such as bounds on the number and maximal norms of indecomposables, are open.

Explicit and minimal criterion sets. I propose to attack criterion sets in the following two ways:

- *Explicit bounds.* Hsia and Icaza [14] made the asymptotic local-global principle of Hsia–Kitaoka–Kneser [15] effective for lattices of rank ≥ 5 , but only when 2 is unramified in K . I am working on removing this hypothesis. The difficulty lies at ramified dyadic primes, where one needs new bounds on value sets of maximal lattices and a neighbour argument that works in the ramified case. If K has enough square classes of totally positive units, universal forms have rank at least 5. For such K this gives an explicit finite criterion set: all elements of $\mathcal{O}_K^+$ of norm up to an explicit bound.
- *Minimal sets.* Running the critical-element test of [18] over this finite set computes C_K exactly. The bounds will be far too large for a naive search, so the practical work is to prune the search space using indecomposables and sails. One goal is obtaining criterion sets for the quadratic family $K = \mathbb{Q}(\sqrt{n^2 - 1})$ with $n^2 - 1$ squarefree, whose only indecomposables up to squares of units are 1 and $\varepsilon = n + \sqrt{n^2 - 1}$. There one expects $C_K = \{1, \varepsilon, 2, 2\varepsilon, \dots, 15, 15\varepsilon\}$ for large n , which would be the first determination of explicit criterion sets for an infinite family of fields. Computing C_K for many small fields, and for other continued-fraction families and the simplest cubic fields, will test this and suggest general patterns.

n -regular lattices (with P. Yatsyna and J. Yoon). A lattice is n -regular if it represents every rank- n lattice that it represents locally everywhere. Oh [32] proved that every n -regular $\mathbb{Z}$ -lattice is n -universal for $n \geq 27$, and gave an exponential lower bound for the minimal rank of n -regular lattices. We plan to generalise both results to totally real fields. Combined with [33], this would give a unified picture of how the minimal ranks of n -universal and n -regular lattices grow.

3 Arithmetic statistics

3.1 Previous work

Unit signature ranks and higher genus theory. The *unit signature rank* of a totally real degree d field K is the $\mathbb{F}_2$ -dimension of the image of the unit group $\mathcal{O}_K^\times$ under the signature map into $\{\pm 1\}^d \cong \mathbb{F}_2^d$. For a real quadratic field $K = \mathbb{Q}(\sqrt{D})$, full unit signature rank means that the negative Pell equation $x^2 - Dy^2 = -1$ is soluble. Koymans and Pagano [25] recently proved Stevenhagen’s conjecture [40] that this holds for approximately 58% of discriminants D with no prime factor $\equiv 3 \pmod{4}$.

With S. H. Man and P. Yatsyna, I study the case of totally real biquadratic fields K . We show that a positive proportion of real biquadratic fields with discriminant $\Delta_K = D^2$, where D has no prime factor $\equiv 3 \pmod{4}$, have all three negative Pell equations soluble, one for each of their three quadratic subfields. Deciding whether K itself has full unit signature rank amounts to deciding whether the narrow 2-class field of K is totally real. This requires classifying all quadratic extensions of K unramified at all finite places (UAFP).

To do this, we extend the higher genus theory of Koymans and Pagano [24] (see also Fröhlich [13]) from the coprime setting to biquadratic fields. When the narrow 4-ranks of the quadratic subfields vanish, this classifies these UAFP extensions through an auxiliary Rédei matrix R' , giving

$$\mathrm{rk}_2 \mathrm{Cl}^+(K) = \omega(K) + \mathrm{corank} R' - 3,$$

where $\omega(K)$ is the number of rational primes ramified in K . It also finds extensions missed by genus theory and a construction of Fouvry and Klüners [12]. We also show that whether each extension is totally real is decided by an explicit sum of Rédei symbols.

Malle’s conjecture. For a finite transitive group $G \leq S_n$, a strong version of a conjecture of Malle [27] predicts that the number of degree- n fields with Galois group G and discriminant at most X grows like $c(G) X^{1/a(G)} (\log X)^{b(G)-1}$ as $X \rightarrow \infty$, for some suitable real constants $a(G), b(G), c(G)$. Klüners [23] showed that Malle’s conjectured b -constant $b_M(G)$ can be wrong. Türkelli [43] proposed a corrected constant $b_T(G)$, and Wang [51] found counterexamples to Türkelli’s version and proposed a further refinement $b_W(G)$ to the b -constant.

With S. Mandal, A. Shnidman, A. Tucker and J. Wang, I developed software that computes all three conjectured b -constants b_M , b_T , and b_W , for transitive groups of degree up to 47, with respect to both the discriminant and the product-of-ramified-primes orderings [28]. Where they disagree, Wang’s b -constant is certified by matching lower bounds from split subextensions with upper bounds from local liftability (including wild pro- p obstructions). The remaining open cases, typically with Q_8 in the kernel (such as $10T20 = C_5^2 \rtimes Q_8$), reduce to a residual extension problem that we treat arithmetically.

3.2 Future work

Unit signature ranks. The next steps are:

- to determine the distribution of $\mathrm{corank} R'$ and prove equidistribution of the Rédei symbols above. When all three quadratic subfields of K have full unit signature rank, K has unit signature rank 3 or 4, and these results should show that a positive proportion of such fields have rank 3. Full rank 4 requires every one of a set of extensions to be real, and this set grows with the number of ramified primes. Numerical evidence of Chan and Milovic [6] suggests each is real independently about half the time, so we conjecture that 100% of these fields, ordered by discriminant, have unit signature rank 3, in sharp contrast with the quadratic case;
- to remove the 4-rank hypothesis, which requires controlling $\mathrm{rk}_8 \mathrm{Cl}^+(K)$ and the cyclic quartic unramified extensions of K ;

- to study fields whose quadratic subfields have prescribed signature ranks, and to extend the method to totally real multiquadratic fields $\mathbb{Q}(\sqrt{d_1}, \dots, \sqrt{d_k})$, where the $2^k - 1$ quadratic subfields all interact; and to consider the analogous questions for totally real cubic fields.

Malle’s conjecture. My first goal is to complete the computation for all transitive groups of degree up to 47, including the remaining Q_8 cases, and to add the results to the LMFDB alongside its number field data. The data will pinpoint exactly the groups where the competing conjectures disagree. These are the groups where a proven asymptotic, or even a matching lower bound, would decide between them, and the smallest such groups are natural targets for new cases of Malle’s conjecture. I also aim to use the data to formulate a group-theoretic characterisation of when b_T and b_W differ, in terms of the subgroup structure and cyclotomic action of G .

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